

AGRIMINE NEXUS

MASTER RESEARCH OPERATING MANUAL (ROM)

Version 1.0

Document Type: Research Governance Manual

Classification: Internal Master Research Standard

1. PURPOSE

This document defines the official research methodology, governance framework, quality standards, evidence requirements, documentation standards, validation process, and deliverables for the AgriMine Nexus project.

It serves as the constitution for all future research activities.

No software architecture, UI/UX design, database design, API development, AI model development, infrastructure planning, or implementation work should begin until the relevant research program has been completed according to this manual.

2. PROJECT VISION

AgriMine Nexus is envisioned as an integrated enterprise operating platform that digitizes and connects Zambia's Agriculture, Mining, Construction, Environmental Engineering, Logistics, Industrial Supply Chains, Water Resources, and Government compliance ecosystems, with long-term scalability across Africa.

The objective is not to build another ERP.

The objective is to build the digital operating system for productive industries.

3. CORE RESEARCH PRINCIPLES

Every research activity shall satisfy the following principles.

Evidence First

Every conclusion must be supported by evidence.

No assumptions.

No speculation.

No hallucinations.

No unsupported recommendations.

Multiple Independent Sources

Every important finding should be validated using multiple independent authoritative sources.

Whenever possible:

Government + Academic + Industry + Practitioner + Company Documentation.

Source Hierarchy

Rank sources according to reliability.

Tier 1

Government publications

Acts

Regulations

National standards

International standards

Official statistics

Government databases

Tier 2

Peer-reviewed journals

Universities

Research institutions

International organizations

Development banks

UN

World Bank

FAO

IFAD

ILO

UNEP

WHO

African Development Bank

Tier 3

Industry associations

Large consulting firms

Annual reports

Technical manuals

Vendor documentation

Conference papers

Professional societies

Tier 4

Public discussions

Community forums

Reddit

Professional blogs

Social media discussions

These sources may reveal user pain points but should never be treated as authoritative evidence without verification.

4. RESEARCH OBJECTIVES

Every research program must answer:

Who performs the work?

Why is the work performed?

When is it performed?

Where is it performed?

How is it performed?

What tools are used?

What regulations apply?

What data is produced?

What data is consumed?

Who owns the data?

What decisions depend on the data?

What risks exist?

What costs exist?

What delays exist?

What software is currently used?

What integrations are required?

What AI opportunities exist?

What automation opportunities exist?

What business opportunities exist?

5. RESEARCH METHODOLOGY

Every research program must follow this lifecycle.

Phase 1

Research Planning

Define objectives

Define scope

Define exclusions

Identify stakeholders

Prepare research questions

Prepare interview questions

Prepare survey questions

Identify datasets

Identify expected outputs

Phase 2

Evidence Collection

Collect documents

Collect regulations

Collect standards

Collect reports

Collect datasets

Collect images

Collect maps

Collect videos

Collect workflows

Collect software screenshots

Collect API documentation

Collect manuals

Collect procurement documents

Collect financial information

Collect organizational structures

Collect public complaints

Collect customer reviews

Collect feature requests

Collect pricing models

Collect integration documentation

Phase 3

Validation

Cross-reference findings

Compare contradictory evidence

Identify outdated information

Identify regional differences

Assess confidence level

Document uncertainties

Phase 4

Knowledge Extraction

Extract entities

Extract business processes

Extract workflows

Extract KPIs

Extract risks

Extract opportunities

Extract software features

Extract integrations

Extract compliance requirements

Extract AI opportunities

Extract GIS opportunities

Extract IoT opportunities

Extract drone opportunities

Phase 5

Product Translation

Transform research into

Business Requirements

Functional Requirements

Non-Functional Requirements

User Stories

Process Models

System Requirements

API Requirements

Database Requirements

Integration Requirements

Security Requirements

AI Requirements

Analytics Requirements

Infrastructure Requirements

Testing Requirements

6. RESEARCH QUALITY CHECKLIST

Every report must include:

Executive Summary

Objectives

Scope

Methodology

Stakeholders

Key Findings

Supporting Evidence

Contradictions

Limitations

Opportunities

Recommendations

Future Research

References

Appendices

7. DATA TO COLLECT FOR EVERY INDUSTRY

Industry Overview

Market Size

Market Value

Growth Rate

Historical Trends

Forecasts

Major Players

Value Chain

Supply Chain

Stakeholders

Business Processes

Daily Operations

Seasonality

Equipment

Assets

Inputs

Outputs

Workforce

Costs

Revenue Streams

Regulations

Licenses

Reporting Requirements

Safety Requirements

Environmental Requirements

Financial Systems

Software Used

Hardware Used

Mobile Usage

Internet Connectivity

Digital Maturity

Pain Points

Automation Opportunities

AI Opportunities

GIS Opportunities

Drone Opportunities

IoT Opportunities

Integration Opportunities

Emerging Technologies

Competitive Landscape

Future Trends

Risk Factors

Success Stories

Failure Cases

Innovation Opportunities

8. INTERVIEW FRAMEWORK

For every stakeholder interview collect:

Background

Role

Organization

Years of Experience

Daily Tasks

Tools Used

Software Used

Manual Processes

Excel Usage

WhatsApp Usage

Paper Forms

Approval Processes

Reporting Processes

Common Delays

Common Errors

Common Costs

Compliance Challenges

Desired Improvements

Technology Adoption

AI Expectations

Integration Needs

Mobile Usage

Offline Needs

Training Needs

Biggest Problems

Dream Solution

9. COMPETITOR RESEARCH STANDARD

For every competitor document:

Company Profile

History

Countries Served

Target Customers

Industry Focus

Architecture

Deployment Model

Technology Stack (if known)

Modules

Features

Pricing

Integrations

Mobile Apps

Offline Support

AI Capabilities

GIS Capabilities

IoT Capabilities

Drone Support

Reporting

Analytics

Security

Compliance

Scalability

Customer Reviews

Strengths

Weaknesses

Missing Features

Innovation Opportunities

Screenshots

Documentation

Videos

Public APIs

Developer Resources

10. TECHNOLOGY RESEARCH STANDARD

Document:

Purpose

Architecture

Advantages

Disadvantages

Licensing

Costs

Scalability

Offline Capability

Performance

Security

Community

Enterprise Adoption

Integration Methods

API Support

SDK Availability

Programming Languages

Deployment Options

Cloud Support

Mobile Support

Future Roadmap

Suitability for Zambia

Suitability for Africa

11. PRODUCT REQUIREMENT EXTRACTION

Every finding must eventually become one or more of the following:

Business Requirement

Functional Requirement

Non-Functional Requirement

User Story

Acceptance Criteria

Database Entity

API Endpoint

Dashboard

Workflow

Notification

Automation Rule

Permission

Security Control

Validation Rule

AI Model

GIS Layer

IoT Integration

External Integration

Report

KPI

12. EVIDENCE SCORING

Every finding should include:

Confidence Score (0 to 100)

Evidence Strength

Source Tier

Publication Date

Geographic Coverage

Industry Coverage

Verification Status

Contradictory Evidence

Research Notes

13. DELIVERABLES FOR EACH RESEARCH PROGRAM

Each completed program must produce:

1. Executive Summary
 2. Detailed Research Report
 3. Stakeholder Analysis
 4. Workflow Diagrams (BPMN where applicable)
 5. Value Chain Map
 6. Pain Point Matrix
 7. Opportunity Matrix
 8. Feature Catalogue
 9. Integration Catalogue
 10. AI Opportunity Matrix
 11. GIS Opportunity Matrix
 12. IoT Opportunity Matrix
 13. Risk Register
 14. Regulatory Checklist
 15. Functional Requirements
 16. Non-Functional Requirements
 17. Data Entity Catalogue
 18. API Candidate List
 19. Dashboard & KPI Catalogue
 20. Research Bibliography
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14. FINAL SUCCESS CRITERIA

Research is considered complete only when it can answer:

• What problem exists? • Who experiences it? • How frequently? • What is the financial impact? • How is it solved today? • Why are current solutions inadequate? • What software exists? • What features are missing? • What regulations must be satisfied? • What integrations are mandatory? • What data is required? • Which KPIs matter? • Which AI capabilities provide measurable value? • Which GIS capabilities provide measurable value? • Which IoT capabilities provide measurable value? • Which workflows should be automated? • What is the minimum viable product? • What is the enterprise roadmap? • What creates a durable competitive advantage?

15. PROJECT PHILOSOPHY

Every recommendation shall be:

Evidence-based.

User-centered.

Regulation-aware.

Technically feasible.

Economically viable.

Scalable.

Secure.

Offline-capable where necessary.

Designed for Zambia first, with an architecture that can scale to Southern Africa and the rest of the continent.

The success of AgriMine Nexus will not be measured by the amount of code written, but by the depth of understanding of the industries it serves and the measurable value it creates for its users.